Part(1): Limits and Sequences

Q1: Find $\lim_{x\to 0^+} \sqrt[x]{x+1}$ $= y \longrightarrow \ln y = \lim_{x\to 0^+} \ln(1+x)^{1/x}$

$= \lim_{x\to 0^+} \frac{\ln(1+x)}{x}$] l'hopital's Rule

$= \lim_{x\to 0^+} \frac{1}{1+x}$

$\ln y = 1.$

$y = e^1 = e.$

Q2: Determine convergence and find limit of $\left\{\left(1+\frac{2}{n}\right)^{2n}\right\}_{n=1}^{\infty}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(1+\frac{2}{n}\right)^{2n} = e^4$

$\rightarrow \lim_{x\to\infty} \left(1+\frac{a}{x}\right)^{bx} = e^{ab}$

$\therefore$ The sequence converges.

Part(2): Infinite Series Convergence Tests

Q3: Test convergence: $\sum_{k=1}^{\infty} \frac{3}{k^5+k}$

$\sum_{k=1}^{\infty} \frac{3}{k^5+k} < \underbrace{\sum_{k=1}^{\infty} \frac{3}{k^5}}_{p=5.}$ by comparision test it converges.

Q4: Test convergence: $\sum_{k=1}^{\infty} \frac{k}{(k^2)!}$

by ratio test: series converges.

$$\lim_{k\to\infty} \left| \frac{a_{k+1}}{a_k} \right| = \left| \frac{k+1}{((k+1)^2)!} \cdot \frac{(k^2)!}{k} \right| = \left| \frac{k+1}{k} \cdot \frac{(k^2)!}{(k+1)^2 (k^2)!} \right| = \lim_{k\to\infty} \frac{1}{k^2+k} = 0 < 1$$

Q5: Test convergence: $\sum_{k=1}^{\infty} \frac{k}{3k^2+1}$

diverges by integral test.

$$\frac{1}{6}\int_1^{\infty} \frac{6x}{3x^2+1}\,dx = \frac{1}{6}\ln[3x^2+1]\Big|_1^{\infty} = \frac{\ln\infty - \ln 4}{6} = \infty$$

(you can also use comparison).

Q6: Test convergence: $\sum_{k=1}^{\infty} \frac{k^2+1}{2k^2+1}$

by divergince test: $\lim_{k\to\infty} \frac{k^2+1}{2k^2+1} = \frac{1}{2} \neq 0$, it diverges.

Q7: Test convergence: $\sum_{k=1}^{\infty} \frac{(k+2)^{\frac{2}{k}}}{3k+1}$, series diverges.

by Ratio test: $\lim_{k\to\infty} \left| \frac{(k+3)^{\frac{2}{k+1}}}{3(k+1)+1} \cdot \frac{3k+1}{(k+2)^{\frac{2}{k}}} \right| = \lim_{k\to\infty} \left| \frac{3k+1}{3k+4} \cdot \frac{(k+3)^{\frac{2}{k+1}}}{(k+2)^{\frac{2}{k}}} \right| = 1$

$$\lim_{k\to\infty} (k+a)^{\frac{b}{k+c}} = e^{b\cdot \lim_{k\to\infty} \frac{\ln(k+a)}{k+c}} = e^0 = 1$$

Part(3): Power Series and Maclaurin Series

Q8: Find interval and radius of convergence for $\sum_{k=1}^{\infty} \frac{(x-3)^k}{\sqrt{k}}$, by Ratio test

- $\lim_{k\to\infty} \left| \frac{(x-3)^{k+1}}{\sqrt{k+1}} \cdot \frac{\sqrt{k}}{(x-3)^k} \right| = |x-3| \lim_{k\to\infty} \sqrt{\frac{k}{k+1}} = |x-3|$

- $|x-3|<1 \longrightarrow -1 < x-3 < 1 \longrightarrow 2 < x < 4$, so interval is $[2,4)$.

↳ $x=4$: $\sum \frac{1}{\sqrt{k}}$ → diverges

↳ $x=1$: $\sum \frac{(-1)^n}{\sqrt{k}}$ → by A.S.T converges

Q9: Find Maclaurin series for $f(x) = x^4 \cos(x^2)$

$$\cos(u) = \sum_{k=0}^{\infty} \frac{(-1)^n u^{2n}}{(2n)!}, \quad u = x^2$$

$$\cos(x)^2 = \sum_{k=0}^{\infty} \frac{(-1)^n x^{4n}}{(2n)!}, \quad f(x) = x^4 \cos(x^2) = \sum_{k=1}^{\infty} \frac{(-1)^n x^{4n+4}}{(2n)!}$$

Part(4): Integration and Improper Integrals

Q10: Evaluate $\int x\sqrt{x^2+4}\,dx$ $= \frac{1}{2}\int 2x\sqrt{x^2+1}\,dx$, $u = x^2+1 \longrightarrow \frac{du}{dx} = 2x$

$$\frac{1}{2}\int \sqrt{u}\,du = \frac{1}{3}u^{3/2} + C = \frac{1}{3}(x^2+1)^{3/2} + C.$$

Q11: Evaluate $\int_3^4 \frac{1}{x^2 - x}\, dx$

$$= -\int_3^4 \frac{-1}{x(x-1)}\, dx = -\int_3^4 \frac{x-1+x}{x(x-1)}\, dx$$

$$= -\int_3^4 \frac{1}{x} - \frac{1}{x-1}\, dx = \ln(x-1) - \ln x \Big]_3^4$$

$$= \ln\left(\frac{x-1}{x}\right)\Big]_3^4 = \ln\frac{3}{4} - \ln\frac{2}{3} = \ln\frac{9}{8}.$$

Q12: Evaluate $\int_0^\infty xe^{-x}\, dx$

u	dv
x	e^{-x}
1	$-e^{-x}$
0	e^{-x}

$$= -e^{-x}(1+x)\Big]_0^a = \frac{x+1}{e^x}\Big]_a^0$$

$$= 1 - \lim_{a\to\infty}\left(\frac{a+1}{e^a}\right) = 1 - \lim_{a\to\infty}\left(\frac{1}{e^a}\right) = 1$$

L'hopital's Rule

Part(5): Polar Coordinates

Q13: Find Cartesian coordinates for $(r, \theta) = (2, \pi)$

$x = r\cos\theta = 2\cos\pi = -2$

$y = r\sin\theta = 2\sin\pi = 0$

$\therefore (-2, 0)$.

Q14: Find polar coordinates for $(x, y) = (\sqrt{5}, \sqrt{5})$

$r = \sqrt{x^2+y^2} = \sqrt{5+5} = \sqrt{10}$

$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}(1) = \pi/4$

$\therefore (\sqrt{10}, \pi/4)$.

Q15: Find Cartesian equation for $r = -2\cos\theta$

$r^2 = -2(r\cos\theta) \longrightarrow x^2+y^2 = -2x$

$x^2+2x+y^2 = 0 \rightarrow (x^2+2x+1)+y^2 = 1$

$\therefore (x+1)^2 + y^2 = 1$.